Structured Output and Schemas

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

Teach layer-specific reasons on complete supplied objects.

Supply trusted source records, keep schemas distinct and use the optional validator within its declared limits.

Keep one habit: Ask which layer first fails and why valid types cannot certify a factual claim.

MAIN LESSON · SECTION 1

Prepare this connected week

> Use one main plan and keep practice separate from the independent first attempt.

Learning goal and precision

Primary target: Separate JSON parsing, **schema validation** and source support checks

For the compact core route, use two 45-minute sessions. All model **output**s and probability tables are constructed classroom examples. No hosted AI service, external API or student account is needed.

Distinguish a model objective from factual support and system permissions. Collect independent work before discussing its solutions.

By the end, I can validate structured **output** in layers: syntax, **schema** and source meaning.

What to prepare

- Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.
- Printed lesson and workbook; paper and pencil
- Calculator; optional local Python 3 interpreter
- Prepare the independent task and fresh retry without displaying solutions

Main teaching plan

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Opening retrieval and readiness

Without opening the old lesson, explain one key idea from Iterative Prompting. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of controlled prompt experiments

Reference example, not the only acceptable wording: Teacher explanation: A prompt experiment compares complete specifications using the same task cases.

Define success before inspecting **output**s; change one intended instruction and preserve other settings. Evaluate both prompts on paired cases with fixed model/version, source records, **output** contract and scoring rubric.

Use development results for selection, then reserve new final cases. Score each case rather than counting attractive sentences.

Without opening the old lesson, explain one key idea from Prompt as a Contract. Give an example and a mistake that the idea helps you avoid.

Look for: Write a prompt specification with observable response criteria

Reference example, not the only acceptable wording: Teacher explanation: **Field**s status,room,capacity,source_ids.

Status supported or needs_review. supported requires matching current approval and current capacity for the same room; no unresolved conflict.

Without opening the old lesson, explain one key idea from A Distribution of Possibilities. Give an example and a mistake that the idea helps you avoid.

Look for: Check whether a next-token table is a valid probability distribution

Reference example, not the only acceptable wording: Teacher explanation: Entries are nonnegative and sum to 1. Not ready has mass 0.3+0.2 = 0.5, including busy and unknown.

No. Parse and validate the returned value, then check its claims and permitted actions against the **evidence** and contract.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Iterative Prompting

Prompt as a Contract

A Distribution of Possibilities

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Schema requires title:text and pages:number. **Output** is {"title":"Cloud","pages":"80"}. Fix its structural error. Would the repaired object prove the book really has 80 pages?

Concept to explain

Week 8: Structured **Output** and **Schema**s

Term	Meaning
schema	A specification of required fields, types and structural rules for data.
field	A named item of information in a record or structured message.

Term	Meaning
type	The category of a value, such as text, number or Boolean.
validation	Checking data or behaviour against rules; in ML, validation data is used to choose settings or models, separately from final testing.

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach the first failing layer without merging the schemas

> Keep room-report and River Notes book fields separately labelled.

Define three layers and the room schema

A response may be valid JSON and still violate the **schema** or sources. First parse the text. Then check the exact object **field**s and **type**s. Finally check source support and domain meaning.

These checks answer different questions. Do not call a successful parser a complete validator.

Use the Week5 room-report contract: exactly status,room,capacity,source_ids. status is supported or needs_review. source_ids is a list of distinct known source-ID strings. For supported, room is a nonempty string and capacity a nonnegative integer. For needs_review, room and capacity are null. Boolean values are not capacities under this contract.

These fragments illustrate failures, not complete candidate objects. A real check receives the whole object and verifies every required **field**.

Do not repair a string capacity automatically unless a separate documented conversion policy permits it; otherwise report **validation** failure. A null value differs from missing a **field**.

Audit the full shared room cards

Card	Complete candidate text
1	{"status":"supported","room":"B","capacity":12,"source_ids":["P","Q"]}
2	{status:"supported","room":"B","capacity":12,"source_ids":["P","Q"]}
3	{"status":"supported","room":"B","capacity":"12","source_ids":["P","Q"]}
4	{"status":"supported","room":"B","capacity":true,"source_ids":["P","Q"]}
5	{"status":"supported","room":"B","capacity":12,"source_ids":["P","Q"],"book":true}
6	{"status":"supported","room":"B","capacity":99,"source_ids":["P","Q"]}

Trace a typed but unsupported capacity

Suppose sources P and Q approve room B and give capacity 12.

The complete object {"status":"supported","room":"B","capacity":99,"source_ids":["P","Q"]} parses and satisfies the stated **field type**s, but its capacity is unsupported by Q. Reject it at **semantic validation**.

Listing a real source ID does not prove that source supports the associated claim.

An object with needs_review and room B violates the conditional **schema** rule even if its source IDs are real. An object with supported, room B and capacity 12 but source ID Z fails the known-source rule.

Require the linked current approval and capacity records to support the same room.

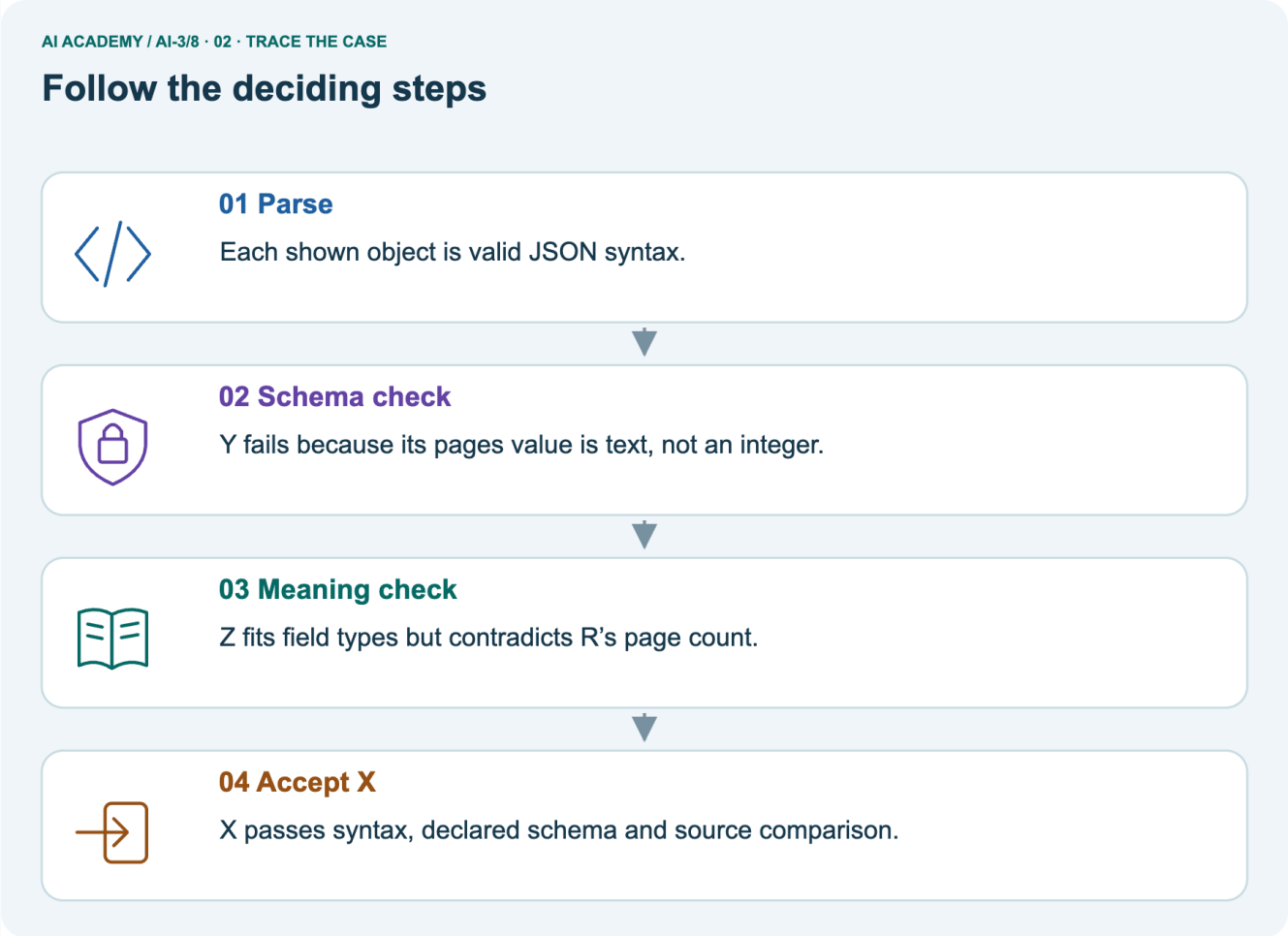
Read the separate book schema and X/Y/Z facts

A fictional Study Buddy returns a machine-readable book fact.

Schema: exactly title text, pages integer, source_id text. Approved record R: River Notes, 64 pages.

X={"title":"River Notes","pages":64,"source_id":"R"}; Y uses pages:"64"; Z uses pages:46 with otherwise identical **field**s.

Trace syntax, type and source checks



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Parse	Each shown object is valid JSON syntax.
2. Schema check	Y fails because its pages value is text, not an integer.
3. Meaning check	Z fits field types but contradicts R's page count.
4. Accept X	X passes syntax, declared schema and source comparison.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Parsing establishes that text forms a structured object. **Schema** checks establish required **field**s and **type**s. **Semantic validation** compares their meaning with the approved source.

These are different layers: a parseable, well-shaped object can still contain a false fact. Report the first failing layer without treating format success as factual success.

Predict the extra rating field

Change: Add an extra **field** "rating":5 to X under the exactly-**field**s **schema**.

Prediction: The changed object fails **schema** despite keeping the correct page count.

Reason: Exactly means additional **field**s are not allowed in this contract.

State source-truth and parser-policy limits

Source agreement does not establish that the approved catalogue itself is correct or complete.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Optional Python reference: the official json documentation describes loads, object_pairs_hook and parse_constant, including default treatment of repeated names and non-finite constants. The classroom example adds its explicit policies. https://docs.python.org/3/library/json.html

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. **Schema field**s: answer=text; **evidence**=list; confidence=low/medium/high.

2. **Output** omits **evidence**: **validation** fails.

3. Confidence=97 is the wrong **type** and allowed-value set.

4. Reject, repair, or request again before downstream use.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

Schema requires title text and pages number. **Output** is {"title":"River","pages":"80"}.

Diagnose the shape error.

Exact explanation to model

Pages is a string, not a number. Use numeric 80. Structure **validation** does not verify the real page count.

Explain the added worked example

Key terms

Term	Meaning
Syntax validation	Checking whether text can be parsed under the format grammar
Schema validation	Checking fields, types and allowed combinations in a parsed value
Semantic validation	Checking meaning and consistency with sources or domain rules
Closed schema	A contract rejecting fields beyond the stated set

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

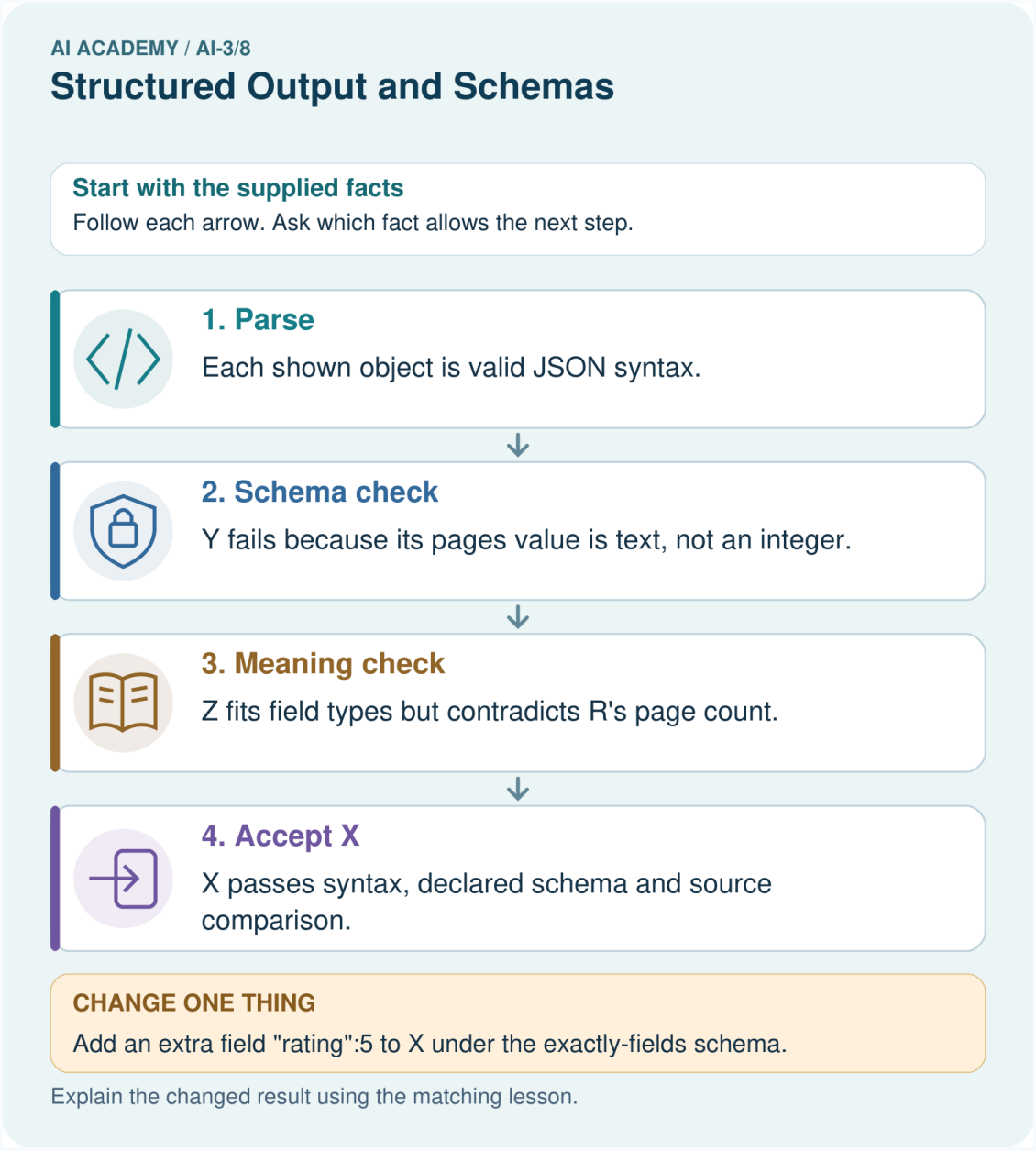
These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Structured Output and Schemas

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

MAIN LESSON · SECTION 3

Guide the explanation, then remove help

> Collect each learner’s reasoning before showing feedback.

Main completion prompt

Completion task: Why do string "12",boolean true,missing capacity and an extra action **field** violate this contract?

Expected reasoning and fading help

Teacher check: Teacher explanation: Capacity must be an integer for supported and null for needs_review, with the **field** present. A Boolean is explicitly excluded.

Extra **field**s violate the **closed schema**; string values are not silently converted.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Structured **output**s and **validation**. Proposed explanation: “Valid JSON means correct answer”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: JSON syntax says nothing about **schema** or source truth.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial question and changed-case reasoning

Which object demonstrates that valid JSON and valid **type**s are insufficient for truth?

Z: its pages is an integer and syntax is valid, but 46 contradicts approved 64.

Changed-case reasoning: The changed object fails **schema** despite keeping the correct page count. Exactly means additional **field**s are not allowed in this contract.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep each learner explaining

- Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.
- Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.
- In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Earlier wording, alternatives and picture transcripts

Compact alternative: this two-session sequence covers selected core tasks. Use the three-session weekly plan for the full lesson, independent transfer, feedback and project work; allow extra time for extensions.

Demonstrate and guide practice

Board sequence

Write the given facts: **Schema** requires title text and pages number. **Output** is {"title":"River","pages":"80"}.

Write the question to explain: Diagnose the shape error.

Reveal after **prediction**s: Pages is a string, not a number. Use numeric 80. Structure **validation** does not verify the real page count.

Use the earlier changed question as optional later practice. Reserve Validate new objects independently for the first independent attempt before hints or feedback.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

Guided: D/8 with P/Q passes syntax, **schema** and source consistency. Zero is a permitted integer but is supported only if Q says zero. Exit: **schema** must reject Boolean capacity and extra **field**s; no action authority exists.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve the prerequisite and distinguish a numerical claim from its interpretation.
7 to 18 minutes	Read the explanation in “Notice the evidence” and “See how it works”, then complete W1; ask for a reason alongside each answer.
18 to 32 minutes	Trace the “A closer explanation” material in “See how it works” and “Follow a worked example”, then complete W2. The pictorial case is an additional example.
32 to 45 minutes	Discuss the diagram and complete W3 in pairs; record support.
Session 2 start to 8 minutes	Complete the guided response before sharing the answer.
8 to 20 minutes	Read the remaining sections and trace another constructed case.
20 to 35 minutes	Collect W4 independently, requiring calculations and a bounded conclusion.
35 to 45 minutes	Discuss W5 and the exit response. Provide feedback and a fresh retry.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the supplied investigation

> Keep expected results, observed paper traces and remaining limits distinguishable.

Supplied materials and answer guidance

Shared cards:1accepted;2syntax;3schema capacity text;4schema Boolean capacity;5closed-schema extra field;6semantic99versusQ12. Missing capacity and unknown Z are schema failures. Legitimate zero is schema-valid but needs matching source evidence. Separate River Notes Xpasses,Ytext64fails schema,Zinteger46fails support;ratingextra fails schema.

Optional program treats repeated names as a declared schema policy and rejects non-JSON constants. Keep the independent F/G task withheld.

Run and record

1. Predict → Test → Record → Explain

2. Keep expected and actual results separate. Surprises are useful data when recorded honestly.

3. Use the six complete candidate strings and trusted P/Q source bundle. Predict syntax, schema and semantic outcomes independently; retain each first failure. The complete local validator is optional and enforces only the declared classroom contract; no external action is performed.

4. What did the hardest test teach you that the normal test did not?

Keep an inspectable record

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose the first misunderstanding

Valid JSON means correct answer: JSON syntax says nothing about **schema** or source truth.

Python treats bool as int, so it is acceptable: The contract explicitly excludes Booleans; an implementation must check accordingly.

A valid report authorises a side effect: Reporting and booking have different permissions.

Choose support from the response

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Make the lab runnable

Use the supplied investigation for shared practice. Keep predictions and observed paper-trace results separate. Reserve Validate new objects independently for the individual first attempt before feedback; do not use the reserved case as a paired lab task.

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the selected new case

> Keep this key until after the first attempt; accept valid alternatives with defensible reasoning.

Selected independent question and key

Selected case: Sources A approve room F and B register F capacity 9. Object X has supported,F,9,[A,B] with exactly the required **field**s. Y replaces 9 with "9". Z keeps integer 9 but names room G.

Trace the first failing layer for each and explain the difference.

Teacher explanation: X passes parsing, **schema** and source **validation**. Y parses but fails **schema** because capacity is a string. Z parses and satisfies **field type**s but fails semantic support because sources name F, not G.

Assuming these values are encoded as proper JSON, syntax is not the failure.

Judge explanation, trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Distinguish the layers

Explain what parsing, **schema** checks and source support each establish.

Teacher explanation: Parsing establishes grammatical JSON; **schema** checks enforce **field**s/**type**s/conditional rules; semantic checks match the claims to sources and domain meaning. None alone establishes all three.

W2 Audit types and fields

Why do string "12",boolean true,missing capacity and an extra action **field** violate this contract?

Teacher explanation: Capacity must be an integer for supported and null for needs_review, with the **field** present. A Boolean is explicitly excluded.

Extra **field**s violate the **closed schema**; string values are not silently converted.

W3 Check a citation-like ID

Sources P,Q support B capacity 12. A **schema**-valid object cites P,Q but gives 99. Which layer rejects it?

Teacher explanation: Semantic/source **validation** rejects unsupported capacity 99. Existing source IDs do not establish entailment.

W5 Keep permissions separate

A fully valid supported reporting object is returned. May the application automatically book the room? Explain.

Teacher explanation: No. The specification is read-only; **validation** of a report does not grant action permission. Booking requires a separately authorised action contract and application controls.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

Pages is a string, not a number. Use numeric 80. Structure **validation** does not verify the real page count.

Additional transfer practice question

A required **schema** has title as text and count as an integer. The **output** is {"title":"Seeds","count":"three"}. Identify the error and give a valid replacement.

Reasoned practice answer

The count value is text instead of an integer. A valid replacement is {"title":"Seeds","count":3}. Valid structure still does not prove factual correctness.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “JSON that looks tidy necessarily satisfies the **schema**.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Use {"title":"Cloud","pages":80}. This satisfies the stated **type**s but the page count still needs a reliable edition-specific source.

Reject malformed or missing **field**s explicitly rather than assuming a successful-looking response is valid.

Explanations for the additional workbook examples

W1 Distinguish the layers

W2 Audit types and fields

W3 Check a citation-like ID

W4 Validate new objects independently

Sources A approve room F and B register F capacity 9. Object X has supported,F,9,[A,B] with exactly the required **field**s. Y replaces 9 with "9". Z keeps integer 9 but names room G.

Trace the first failing layer for each and explain the difference.

Teacher explanation: X passes parsing, **schema** and source **validation**. Y parses but fails **schema** because capacity is a string. Z parses and satisfies **field type**s but fails semantic support because sources name F, not G.

Assuming these values are encoded as proper JSON, syntax is not the failure.

W5 Keep permissions separate

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, adapt and return

> Retain the project evidence and revisit the reasoning on a changed case.

Project evidence to keep

Require answer, source snippet, citation, uncertainty, and next-question **field**s.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Validate Study Buddy **output**s before another program consumes them; retain source IDs and separate syntax/**type**/support failures.

Support while preserving the reasoning goal

Provide one short approved passage; separate instruction, **input**, **output** and **evidence**. Stretch: introduce an unsupported claim or conflicting source and validate claim by claim with a fallback.

Provide labelled intermediate columns and allow a calculator. Ask students to state the meaning and denominator of a value before interpreting it.

Extension: implement the classroom calculation locally and test boundary cases without using personal data.

W4 requires X passes all, Y **schema** failure, Z semantic failure. Fresh retry: sources approve H capacity 0.

Object H/0 passes; H/null with supported fails **schema**; H/1 with integer capacity passes **schema** but fails semantic support. Preserve legitimate zero.

Plan the delayed check

Suggested return points in this level: ai-3/9, ai-3/11, ai-3/14. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	

Record	Teacher evidence
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge to next week

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 9: Prompt Evaluation. Retrieve today's **evidence** and identify what carries forward and what is new.

Next is Week9: Prompt Evaluation. Carry forward explicit syntax, schema and source-support criteria when comparing fixed development cases and protecting reserved final evidence.

▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

Verified Study Buddy

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference: alternative planning checklist**

REFERENCE · SECTION 7

Reference: alternative planning checklist

> Use as a reminder alongside the main route rather than teaching the cycle twice.

Preparation and schedule

Teaching target: Separate JSON parsing, **schema validation** and source support checks

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
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> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
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> **Session:** Session 3 · Investigate and transfer · 45 min
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> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Practice and assessment reminders

Before class, solve the supported practice: Why do string "12",boolean true,missing capacity and an extra action **field** violate this contract?

Reasoned practice answer: Teacher explanation: Capacity must be an integer for supported and null for needs_review, with the **field** present. A Boolean is explicitly excluded.

Extra **field**s violate the **closed schema**; string values are not silently converted.

Check every learner before discussion: Use the worked case for Structured **output**s and **validation**. Proposed explanation: “Valid JSON means correct answer”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: JSON syntax says nothing about **schema** or source truth.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Add an extra **field** "rating":5 to X under the exactly-**field**s **schema**. Require a **prediction** and a reason before comparing. Expected result: The changed object fails **schema** despite keeping the correct page count.

Reason: Exactly means additional **field**s are not allowed in this contract.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference: feedback and records**

REFERENCE · SECTION 8

Reference: feedback and records

> Use digital tools where they help retain an explanation and its evidence.

Keep practice and independent evidence separate

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Repair and portfolio records

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.
</details>

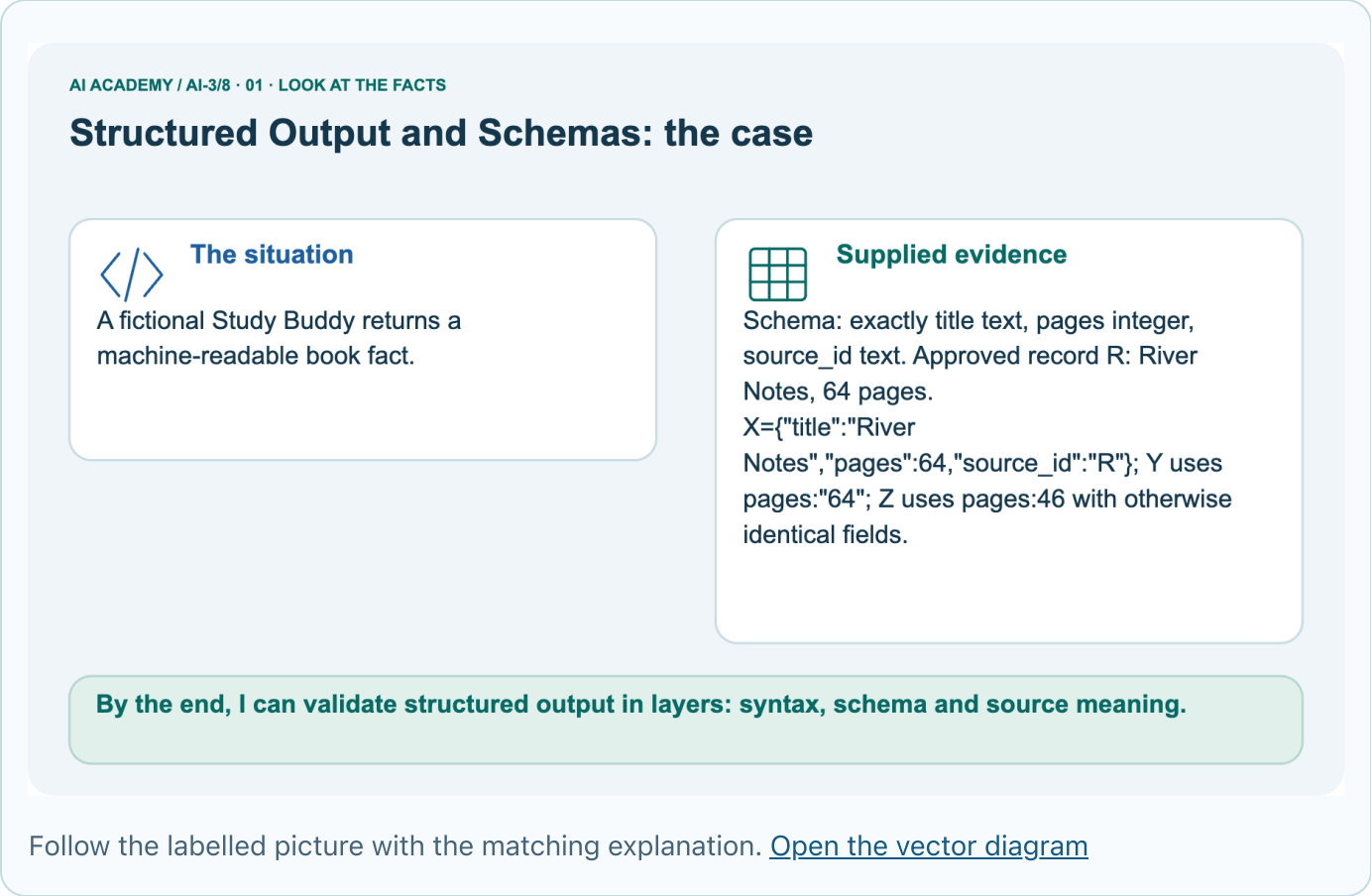
<details>
<summary>Reference: teach with the six pictures</summary>
</details>

REFERENCE · SECTION 9

Reference: teach with the six pictures

> Choose the picture for the teaching move and connect it to the explanation.

01. Structured Output and Schemas: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

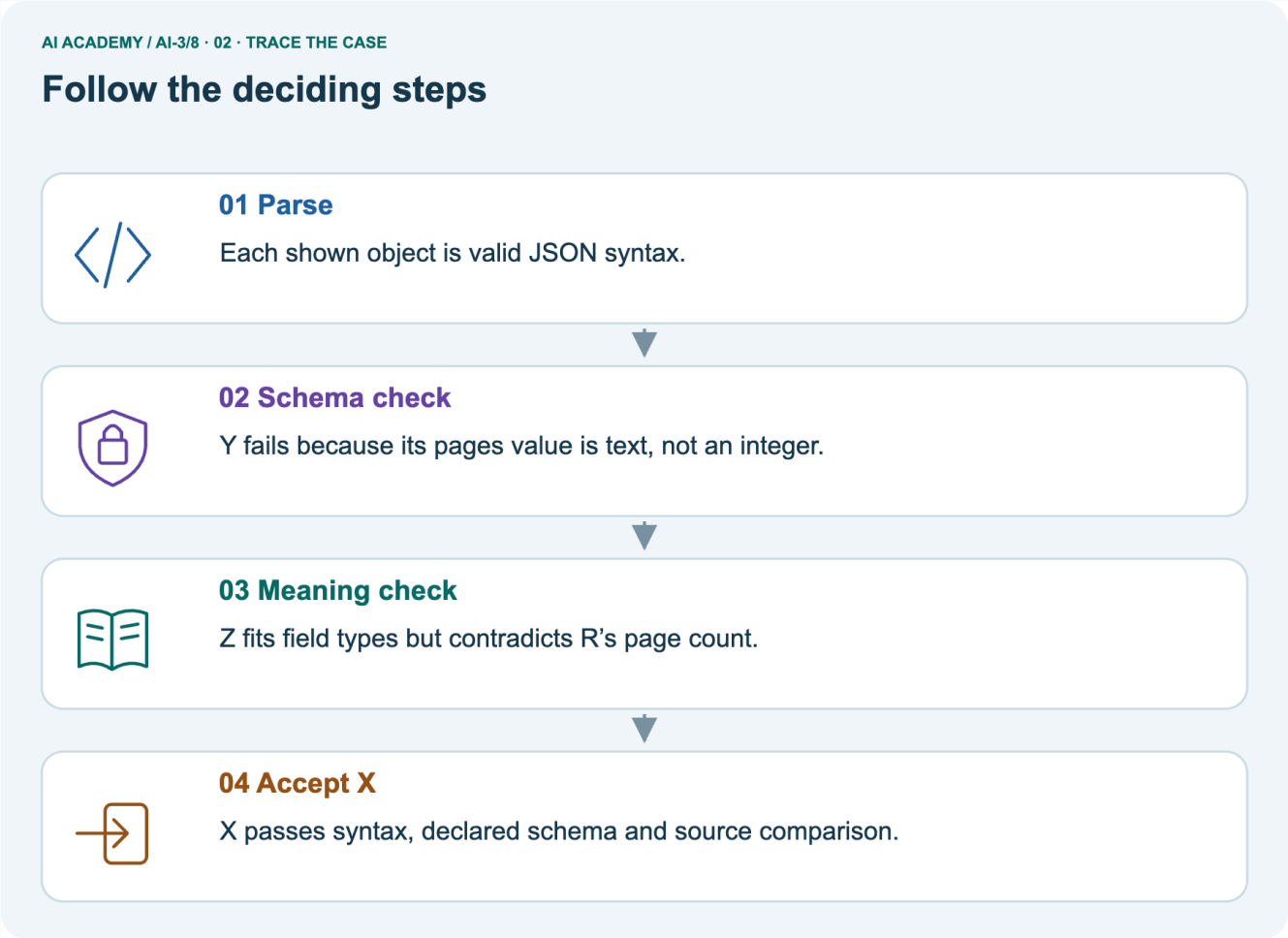
A fictional Study Buddy returns a machine-readable book fact. Supplied facts: **Schema**: exactly title text, pages integer, source_id text. Approved record R: River Notes, 64 pages.

X={"title":"River Notes","pages":64,"source_id":"R"}; Y uses pages:"64"; Z uses pages:46 with otherwise identical **field**s.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



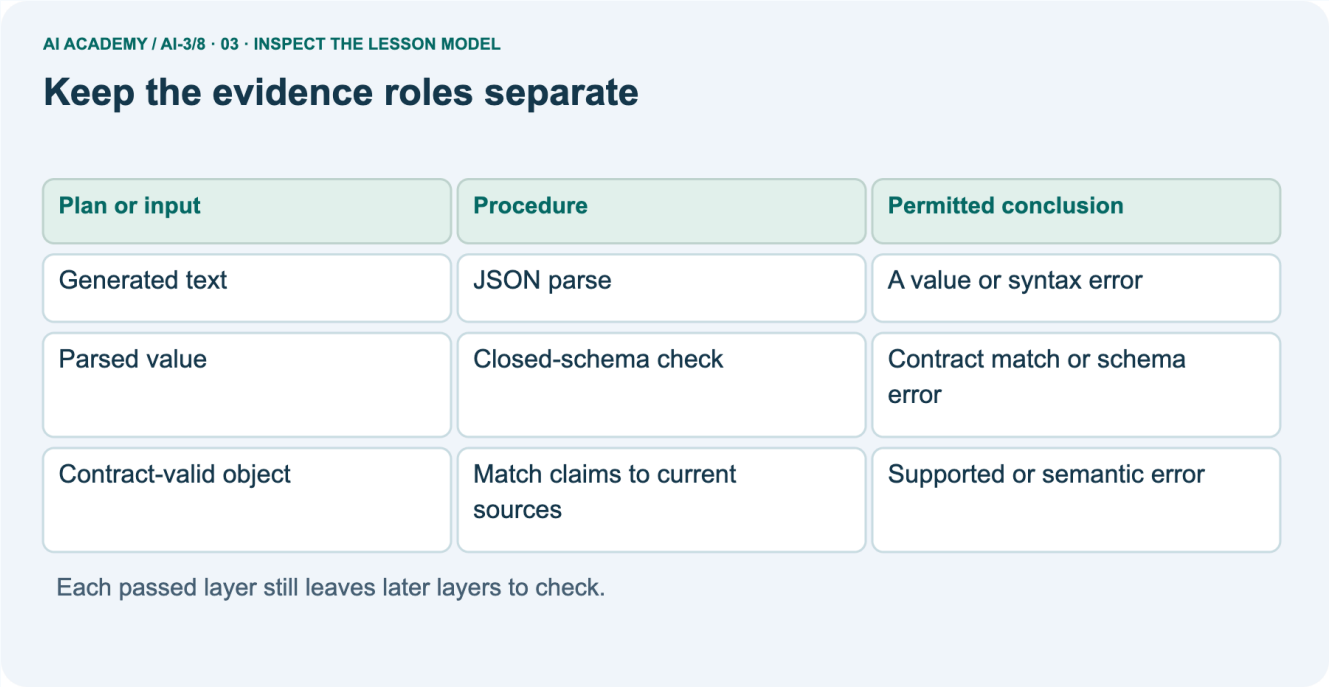
Follow the labelled picture with the matching explanation. Open the vector diagram

Parse: Each shown object is valid JSON syntax. → **Schema** check: Y fails because its pages value is text, not an integer. → Meaning check: Z fits **field type**s but contradicts R's page count. → Accept X: X passes syntax, declared **schema** and source comparison.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Keep the evidence roles separate



Follow the labelled picture with the matching explanation. Open the vector diagram

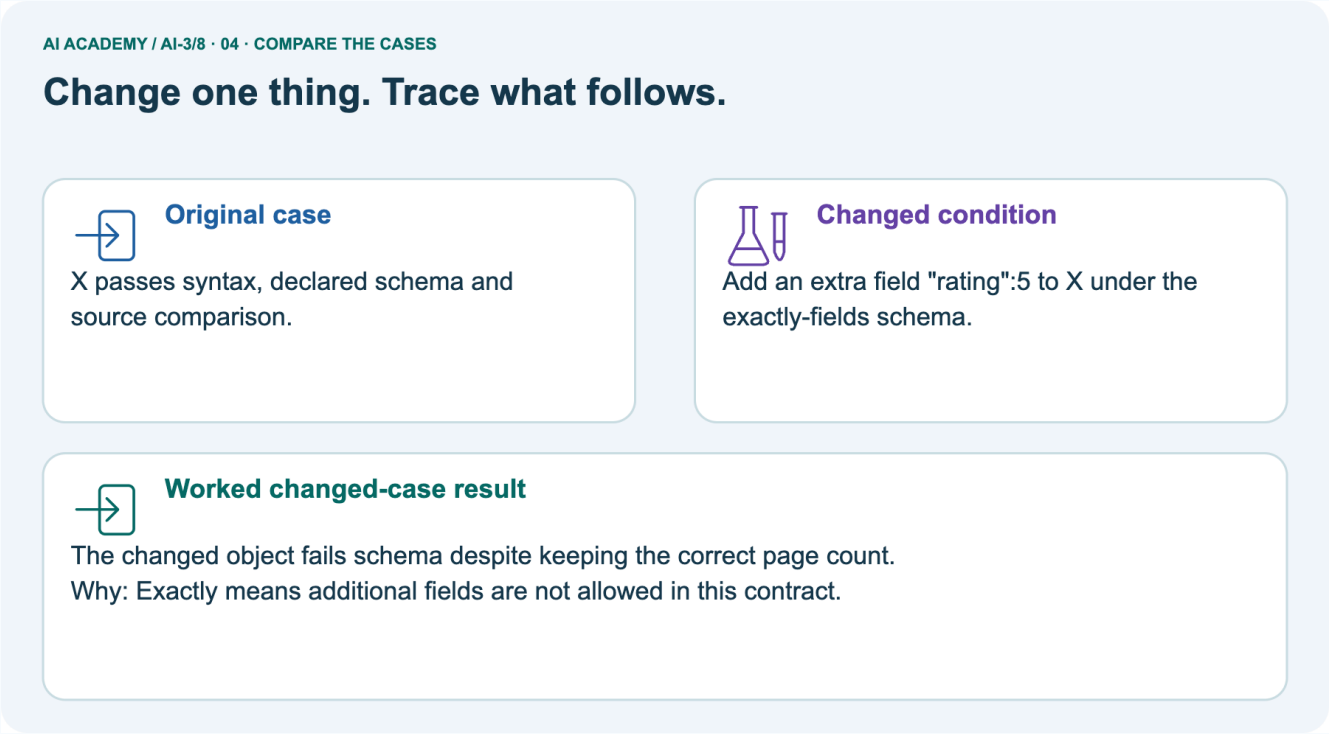
Keep the **evidence** roles separate. Generated text | JSON parse | A value or syntax error. Parsed value | Closed-**schema** check | Contract match or **schema** error.

Contract-valid object | Match claims to current sources | Supported or semantic error

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: X passes syntax, declared **schema** and source comparison. Changed condition: Add an extra **field** "rating":5 to X under the exactly-**field**s **schema**.

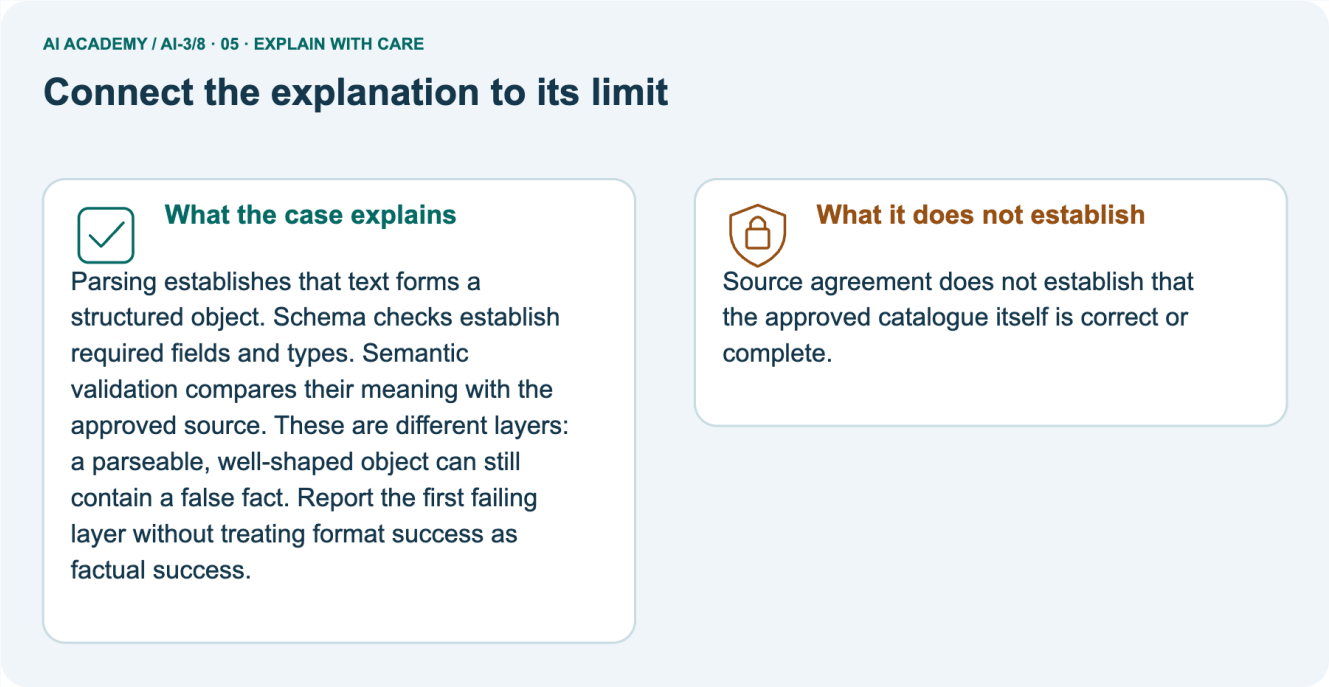
Expected result: The changed object fails **schema** despite keeping the correct page count. Reason: Exactly means additional **field**s are not allowed in this contract.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Exactly means additional **field**s are not allowed in this contract.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Parsing establishes that text forms a structured object. **Schema** checks establish required **field**s and **type**s. **Semantic validation** compares their meaning with the approved source.

These are different layers: a parseable, well-shaped object can still contain a false fact. Report the first failing layer without treating format success as factual success.

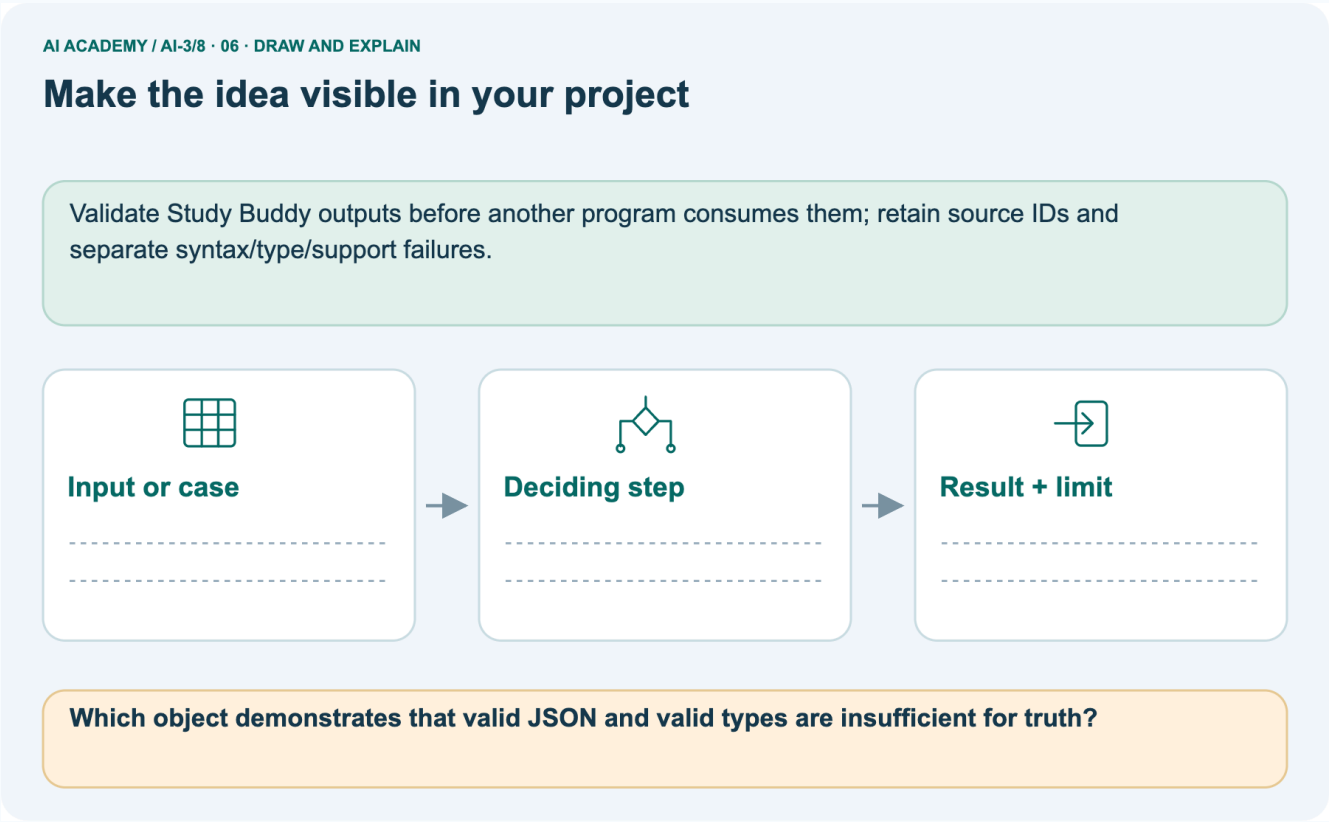
Limit: Source agreement does not establish that the approved catalogue itself is correct or complete.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Source agreement does not establish that the approved catalogue itself is correct or complete.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Validate Study Buddy **output**s before another program consumes them; retain source IDs and separate syntax/**type**/support failures. Draw your **input**, deciding step, result and limit.

Supported question: Which object demonstrates that valid JSON and valid **type**s are insufficient for truth?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Z: its pages is an integer and syntax is valid, but 46 contradicts approved 64.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Structured Output and Schemas: the case

02. Follow the deciding steps

03. Keep the evidence roles separate

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.